

NEON SHORTS ENGINE

Autonomous Procedural Vertical Video Factory for YouTube Shorts, TikTok & Reels

RESOLUTION: 1080x1920 (9:16)
PIPELINE: WebCodecs + DSP Audio
TEST SUITE: 15/15 PASSING (100%)
ENGINE VERSION: 1.0.0 PRO

1. SYSTEM ARCHITECTURE & CORE PIPELINE

🚀 Dual-Pass Render

Pass 1: Pre-calculates physics & DSP sound in pure headless JS (zero UI lag).

Pass 2: Renders frames to OffscreenCanvas and streams directly to GPU H.264 encoder.

🔊 Pure Mathematical DSP

Zero external MP3/WAV files. Real-time synthesis on Float32Array planar memory. Pentatonic ASMR scales, Tanh limiter, zero-click fades.

🌐 5-Language Global Reach

TR, EN, ES, PT, DE. Localized studio UI, in-game HUD, and automated anti-spam metadata generator (Day_XXX_info.txt).

2. QUICK START & WORKFLOW IN 4 STEPS

STEP 1

Launch Studio

```
npm install
npm run dev
```

Navigate to localhost:5173 in Chrome or Brave.

STEP 2

Select Mode & Lang

Choose from 25 modes on the left. Pick target language and teams/rivalries.

STEP 3

Authorize Output

Click [ Select Directory] to grant automatic local file write access.

STEP 4

Export Video (MP4)

Click Single Export or [ 7-Day Series]. Render takes ~15 seconds!

3. VIRAL GAME MODES CATALOG (PART 1: MODES 01 – 09)

01	🗡️ Territory War	700+ grid territorial clash between nations, rivalries, or follower names. High comment velocity.	GRID PHYSICS
02	🔢 Multiplier Gates	Balls pass through x2, x3, and +25 multiplier gates, cascading into a massive Jackpot waterfall.	CASCADE JACKPOT
03	🌀 Hexagon Escape	5 concentric counter-rotating neon hexagons with narrow escape slots. Hypnotic retention driver.	ASMR MAZE
04	🔴 Circle Escape	Accelerating ball chipping away at breakable neon perimeter segments. High-tension climax.	SHATTER PHYSICS
05	🎰 Plinko Cascade	120+ brass pegs with multiplying collector bins. Satisfying audio chimes on every bounce.	PEGS & BINS
06	💀 Battle Royale	Shrinking electric storm ring with saw obstacles. Eliminates balls until one champion remains.	LAST SURVIVOR
07	🏰 Tower Crush	Mega trampoline launching wrecking balls to demolish tiered neon blocks with explosive bass.	DESTRUCTION
08	🕳️ Black Hole	Rocket-thruster particles battling intense gravitational pull and event horizon spaghettification.	ORBITAL DECAY
09	🏔️ Stair Race	32 vertical pyramid stairs equipped with bouncy trampolines racing to touch the summit trophy.	VERTICAL CLIMB

3. VIRAL GAME MODES CATALOG (PART 2: MODES 10 – 25)

10	 Laser Crossfire	Pulsing and rotating lethal laser beams across the arena. Particles vaporize on contact with spark bursts.	LASER VAPORIZE
11	 Sawblade Gauntlet	High-speed spinning neon buzzsaws arranged vertically. Balls navigate narrow gaps to reach the safety exit.	BUZZSAW DESC
12	 Domino Cascade	220-tile continuous spiral domino collapse. Rhythmic ASMR topples leading to a golden gong finish.	CHAIN REACTION
13	 Pendulum Hammers	Physically oscillating massive pendulum hammers swinging horizontally. Heavy bass impact audio cues.	HARMONIC SWING
14	 Tug of War	Two opposing teams clash against bumpers, pulling a central glowing energy knot across boundary thresholds.	CORE TENSION
15	 Time Bomb	Hot-potato ticking time bomb transferred upon contact. Decreasing fuse timers trigger explosive eliminations.	HOT POTATO
16	 Wall Climb	Equal 340px vertical corridors. Balls bounce zig-zag upwards with upward velocity kicks to ring the summit bell.	ZIG-ZAG BOUNCE
17	 Ice vs Lava	Frost vs Magma elemental tug-of-war. Balanced clashes dynamically shift the thermal biome divide line.	THERMAL CLASH
18	 Cell Mitosis	Geometric prism splitters multiply cells exponentially. Pentatonic scale chime progression creates ASMR bliss.	EXPONENTIAL
19	 Magnetic Polarity	Dual rotating magnetic poles (+/-) with Coulomb forces, mutual spark discharge, and dynamic electric arcs.	COULOMB FORCE
20	 Pachinko Madness	Authentic Tokyo pachinko field with responsive tulip cups, metallic pin acoustics, and Fever Jackpot frenzy.	FEVER JACKPOT
21	 Portal Paradox	Momentum-conserving paired wormholes. Balls build slingshot speed through infinite inter-dimensional loops.	WORMHOLE SLING
22	 Gravity Inversion	Gravity vector flips 90 degrees every 3.0 seconds (Down → Right → Up → Left). Neon perimeter trampolines.	VECTOR SHIFT
23	 Archimedes Spiral	Conservation of angular momentum: racers accelerate exponentially as they spiral toward the central vortex core.	VORTEX INWARD
24	 Cyber Pinball	Auto-triggered mechanical flippers, neon bumpers with score multipliers, and drain reset mechanics.	ARCADE FLIPPER
25	 Helix Fall	3D rotating cylinder drop with gap alignments. Consecutive drops ignite Fireball mode, shattering tower floors.	FIREBALL SMASH

💡 VIRAL ENGAGEMENT SECRET: "THE RIVALRY MULTIPLIER"

PRO STRATEGY

Modes like **Territory War**, **Tug of War**, and **Ice vs Lava** trigger emotional tribalism (e.g. **TR** vs **BR** or Real Madrid vs Barcelona). Viewers passionately defend their team in the comments, triggering the YouTube Shorts / TikTok algorithm to classify the video as "Highly Engaging" and push it to millions of new feeds.

4. 9:16 VERTICAL SAFEZONE & AUDIO DSP SPECS

TOP HEADER / SEARCH / LIVE BADGES (Y: 0 - 160)

870 x 1360 PIXELS
ACTIVE COLLISION & HUD ARENA
X: 70px to 940px • Y: 160px to 1520px

BOTTOM CAPTION / SOUND / SHARE BUTTONS (Y: 1520 - 1920)

Mathematical Audio Synthesis

- ▶ **Zero-Click Envelopes:** 40ms micro fade-in and fade-out prevents speaker pops and ear fatigue.
- ▶ **Soft Tanh Limiter:** Equation $y = \tanh(x)$ guarantees signals never exceed $[-1.0, 1.0]$, preventing harsh clipping during 50+ ball collisions.
- ▶ **Pentatonic Tuning:** A4 root harmonic frequencies (220 Hz to 880 Hz) naturally create soothing ASMR chime progressions.

5. BATCH PRODUCTION & METADATA FACTORY

Multi-Worker Pool

Web Worker runs independently of the browser UI. The studio interface maintains silky smooth 60 FPS while exporting complex 1080×1920 videos.

7-Day Automation

[⚡ GENERATE 7-DAY PACK] creates 7 distinct seed simulations in sequence. Produce an entire week of content in under 2 minutes.

SEO & Hook Generator

Each video outputs Day_XXX_info.txt containing localized viral hook titles, high-CTR descriptions, and platform-specific hashtags.

6. VIRAL ALGORITHM OPTIMIZATION RULES

1. The 3-Second Hook Velocity

Simulations start immediately with fast-moving balls, vibrant neon sparks, and crisp audio to eliminate swipe-away rate within the first 3 seconds.

2. Perfect Looping & 110%+ Retention

Videos conclude at second 28 with a decisive victory screen. The clean cut seamlessly flows back into the start, causing repeat views that skyrocket reach.

3. Comment War Call-to-Action

Pin this comment on every upload: *"Which country should join tomorrow's battle? Comment below!"* Comment count is the #1 Shorts ranking signal.

4. Upload Timing Protocol

Post twice daily: Slot 1 between 12:00 PM – 2:00 PM (lunch commute), Slot 2 between 7:00 PM – 9:30 PM (peak evening entertainment).

7. VERIFICATION & TEST SUITE

\$ npm test

✓ i18n - 5 Language Integrity & 25 Mode Dictionaries
✓ DynamicMetadata - Multi-Language Metadata Generation
✓ DSPSoundSynth - Tanh Soft Limiter $[-1.0, 1.0]$ Safety

i tests 15 | pass 15 | fail 0 | status: 100% PRODUCTION READY

✓ ModeManager - Frame 0 Render & 120-Step Simulation
✓ PRNG - LCG Seed Determinism & Distribution
✓ SafeZone - 9:16 Mathematics & UI Padding